WORKED SOLUTION

STEP II 1988, Question 2

Pure • Symmetric Algebraic Equations

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/88-s2-q2

The key idea

Clear the fractions, then subtract one equation from the other. The difference of two squares on the right produces a factor $y - z$, matching the $y - z$ that appears on the left, and the hypothesis $y \neq z$ is exactly what lets us cancel it. After that, everything comes down to one symmetric expression in x, y and z, and symmetry does the rest.

Solution

Proving $x + y + z = a$

Since $y \neq 0$ and $z \neq 0$ we may multiply through:

$$2ay - 3y^2 = (z - x)^2, \tag{1}$$
$$2az - 3z^2 = (x - y)^2. \tag{2}$$

Subtracting (2) from (1), and using a difference of two squares on the right,

$$\begin{aligned} 2a(y - z) - 3(y^2 - z^2) &= (z - x)^2 - (x - y)^2 \\ &= [(z - x) - (x - y)][(z - x) + (x - y)] \\ &= (y + z - 2x)(z - y). \end{aligned}$$

The left-hand side is $(y - z)[2a - 3(y + z)]$, so

$$(y - z)[2a - 3(y + z)] = -(y - z)(y + z - 2x).$$

Ask yourself Where exactly is $y \neq z$ needed?

Right here: because $y - z \neq 0$ we can divide by it, giving

$$2a - 3y - 3z = 2x - y - z \quad \Longrightarrow \quad 2a = 2x + 2y + 2z,$$

so $x + y + z = a$, as required.

Proving the third equation

Substitute $a = x + y + z$ back into (1):

$$y(2x + 2y + 2z - 3y) = (z - x)^2 \implies 2xy - y^2 + 2yz = z^2 - 2zx + x^2,$$

which rearranges to

$$x^2 + y^2 + z^2 - 2xy - 2yz - 2zx = 0. \tag{3}$$

The striking thing is that (3) is *symmetric* in x, y and z. Now look at what we want. Since $x \neq 0$, the equation $2a - 3x = \dfrac{(y-z)^2}{x}$ is equivalent to

$$x(2a - 3x) = (y - z)^2,$$

and with $a = x + y + z$ this reads

$$x(2y + 2z - x) = y^2 - 2yz + z^2 \iff x^2 + y^2 + z^2 - 2xy - 2yz - 2zx = 0,$$

which is exactly (3). Hence

$$2a - 3x = \frac{(y-z)^2}{x}.$$

(In fact all three equations are just (3) together with $a = x + y + z$, each written from the point of view of one variable.)

Can the last equation hold when $y = z$?

Yes. If $y = z$ the two given equations become the same equation, $2a - 3y = \dfrac{(y-x)^2}{y}$, and the last equation becomes $2a - 3x = 0$, i.e. $a = \frac{3}{2}x$. Putting $a = \frac{3}{2}x$ into the first,

$$3x - 3y = \frac{(x-y)^2}{y} \iff 3y(x - y) = (x - y)^2 \iff (x - y)(x - 4y) = 0.$$

So, for instance, $x = y = z = 1$ with $a = \frac{3}{2}$ satisfies all three equations:

$$2a - 3y = 0 = \frac{(z-x)^2}{y}, \qquad 2a - 3z = 0 = \frac{(x-y)^2}{z}, \qquad 2a - 3x = 0 = \frac{(y-z)^2}{x}.$$

Another example is $x = 4$, $y = z = 1$, $a = 6$: then $2a - 3y = 9 = \frac{(1-4)^2}{1}$ for each of the first two, and $2a - 3x = 0 = \frac{0^2}{4}$.

So the last equation does *not* hold only if $y \neq z$: the condition $y \neq z$ guarantees it, but it can also hold when $y = z$. What fails when $y = z$ is the *guarantee*. For example $x = 1$, $y = z = 2$, $a = \frac{13}{4}$ satisfies both given equations (each side equals $\frac{1}{2}$), but $2a - 3x = \frac{7}{2} \neq 0$. Notice too that the first example has $x + y + z = 3 \neq a$, so with $y = z$ even the first conclusion can fail.

Checks

- Take $x = 9$, $y = 4$, $z = 1$, which satisfy (3): $81 + 16 + 1 - 72 - 8 - 18 = 0$. Then $a = 14$ and $2a - 3y = 16 = \frac{(1-9)^2}{4}$, $2a - 3z = 25 = \frac{(9-4)^2}{1}$ and $2a - 3x = 1 = \frac{(4-1)^2}{9}$. All three hold.
- A numerical check verified the identity $(1) - (2) = (y - z)\big(2a - 2(x + y + z)\big)$ exactly on 10,000 random rational triples, and confirmed all three equations on about 1,900 random points of (3) with $y \neq z$, plus the three $y = z$ examples above.

Common slips

- Dividing by $y - z$ without saying why it's allowed. That step is the whole reason the hypothesis is there.
- Answering the final part with "no" or "yes" and no example. A single explicit triple with $y = z$ settles it.
- Thinking that $y = z$ makes the last equation false. It only removes the guarantee.

Beyond STEP

Equation (3) is a quadratic form, and its zero set is a double cone in (x, y, z)-space. If $x, y, z > 0$, write $x = p^2$, $y = q^2$, $z = r^2$: then (3) factorises as

$$-(p + q + r)(-p + q + r)(p - q + r)(p + q - r) = 0,$$

which is 16 times the squared area of a triangle with sides p, q, r (Heron's formula). So (3) says that $\sqrt{x}$, $\sqrt{y}$, $\sqrt{z}$ form a degenerate triangle, for example $\sqrt{9} = \sqrt{4} + \sqrt{1}$ in the check above. Quadratic forms and their classification are a standard topic in first-year linear algebra; see Strang, *Introduction to Linear Algebra*.

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